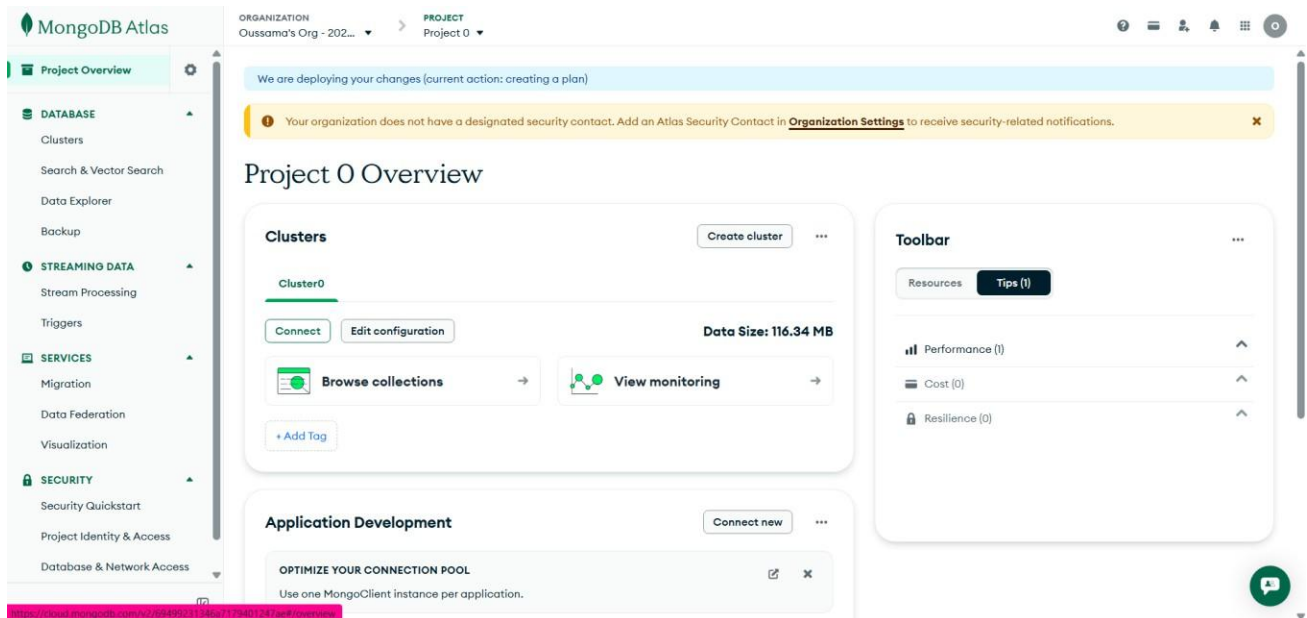


Lab 4: Initiation à MongoDB

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2ème Année Cycle d'Ingénieur GI
2025-2026
BIG DATA

1. Création d'un compte et déploiement d'un cluster



2. Connexion à MongoDB Atlas

```
PS C:\Users\oussa> mongosh --version or mongo --version
2.5.10
PS C:\Users\oussa> mongosh "mongodb+srv://cluster0.rwja68.mongodb.net/" --apiVersion 1 --username oukhmir_oussama
Enter password: *****
Current Mongosh Log ID: 6949a57c16df987b671e2620
Connecting to:      mongodb+srv://<credentials>@cluster0.rwja68.mongodb.net/?appName=mongosh+2.5.10
Using MongoDB:      8.0.17 (API Version 1)
Using Mongosh:       2.5.10

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

Atlas atlas-jh69i3-shard-0 [primary] test> |
```

3. Création d'une base de données et d'une collection

- ✓ Pour se placer dans une base
- ✓ Créer une collection nommée users
- ✓ Lister les collections de la base
- ✓ Insérer un document JSON dans la collection users
- ✓ Insérer maintenant plusieurs tuples dans la collection users

```
mongosh mongodb+srv://<a> x + v
Enter password: *****
Current Mongosh Log ID: 6949a57c16df987b671e2620
Connecting to: mongodb+srv://<credentials>@cluster0.rwjga68.mongodb.net/?appName=mongosh+2.5.10
Using MongoDB: 8.0.17 (API Version 1)
Using Mongosh: 2.5.10

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

Atlas atlas-jh69i3-shard-0 [primary] test> show db
MongoshInvalidInputError: [COMMON-10001] 'db' is not a valid argument for "show".
Atlas atlas-jh69i3-shard-0 [primary] test> show dbs
sample_mflix 114.65 MiB
admin 360.00 MiB
local 3.02 GiB
Atlas atlas-jh69i3-shard-0 [primary] test> use BankDB
switched to db BankDB
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.createCollection("users")
{ ok: 1 }
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.insertOne({ name: "Mohamed", age: 20, city: "Casablanca" })
{
  acknowledged: true,
  insertedId: ObjectId('6949a94516df987b671e2621')
}
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.insertMany([
... { name: "Alice", age: 25, city: "Paris" },
... { name: "Bob", age: 30, city: "Lyon" },
... { name: "Charlie", age: 28, city: "Marseille" }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6949a94c16df987b671e2622'),
    '1': ObjectId('6949a94c16df987b671e2623'),
    '2': ObjectId('6949a94c16df987b671e2624')
  }
}
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

4. Requêtes pour récupérer des données

- ✓ L'objet (javascript) implicite, db, permet de soumettre des demandes d'exécution de certaines méthodes. Afficher tous les utilisateurs de la collection:

```
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.insertMany([
...   { name: "Alice", age: 25, city: "Paris" },
...   { name: "Bob", age: 30, city: "Lyon" },
...   { name: "Charlie", age: 28, city: "Marseille" }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6949a94c16df987b671e2622'),
    '1': ObjectId('6949a94c16df987b671e2623'),
    '2': ObjectId('6949a94c16df987b671e2624')
  }
}
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.find()
[
  {
    _id: ObjectId('6949a94516df987b671e2621'),
    name: 'Mohamed',
    age: 20,
    city: 'Casablanca'
  },
  {
    _id: ObjectId('6949a94c16df987b671e2622'),
    name: 'Alice',
    age: 25,
    city: 'Paris'
  },
  {
    _id: ObjectId('6949a94c16df987b671e2623'),
    name: 'Bob',
    age: 30,
    city: 'Lyon'
  },
  {
    _id: ObjectId('6949a94c16df987b671e2624'),
    name: 'Charlie',
    age: 28,
    city: 'Marseille'
  }
]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

- ✓ Compter le nombre de documents dans la collection et Récupérer un utilisateur spécifique :

```
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.count()
DeprecationWarning: Collection.count() is deprecated. Use countDocuments or estimatedDocumentCount.
4
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.findOne({ name: "Alice" })
{
  _id: ObjectId('6949a94c16df987b671e2622'),
  name: 'Alice',
  age: 25,
  city: 'Paris'
}
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.find({ age: { $gt: 20 } })
[
  {
    _id: ObjectId('6949a94c16df987b671e2622'),
    name: 'Alice',
    age: 25,
    city: 'Paris'
  },
  {
    _id: ObjectId('6949a94c16df987b671e2623'),
    name: 'Bob',
    age: 30,
    city: 'Lyon'
  },
  {
    _id: ObjectId('6949a94c16df987b671e2624'),
    name: 'Charlie',
    age: 28,
    city: 'Marseille'
  }
]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.updateOne({ name: "Alice" }, { $set: { age: 26 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

- ✓ Filtrer les utilisateurs par âge (age>20) et Modifier l'âge d'un utilisateur et Supprimer un utilisateur et supprimer une collection:

```
mongosh mongodb+srv://<cc> x + v
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.findOne({ name: "Alice" })
{
  _id: ObjectId('6949a94c16df987b671e2622'),
  name: 'Alice',
  age: 25,
  city: 'Paris'
}
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.find({ age: { $gt: 20 } })
[
  {
    _id: ObjectId('6949a94c16df987b671e2622'),
    name: 'Alice',
    age: 25,
    city: 'Paris'
  },
  {
    _id: ObjectId('6949a94c16df987b671e2623'),
    name: 'Bob',
    age: 30,
    city: 'Lyon'
  },
  {
    _id: ObjectId('6949a94c16df987b671e2624'),
    name: 'Charlie',
    age: 28,
    city: 'Marseille'
  }
]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.updateOne({ name: "Alice" }, { $set: { age: 26 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.deleteOne({ name: "Charlie" })
{ acknowledged: true, deletedCount: 1 }
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.users.drop()
true
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

5. Importation de fichier dans une collection

- ✓ Télécharger le fichier transactions.json. et Après avoir créé la base de données appelée « BankDB » et la collection « transactions », importer le fichier dans la collection

```
Administrateur : Windows Po x + v
PS C:\Users\oussa\Documents> mongoimport --uri "mongodb+srv://oukhmir_oussama@cluster0.rwjga68.mongodb.net/BankDB" --collection transactions --file transact
ions.json --jsonArray
Enter password for mongo user:
2025-12-22T21:43:17.214+0100 connected to: mongodb+srv://[**REDACTED**]@cluster0.rwjga68.mongodb.net/BankDB
2025-12-22T21:43:19.959+0100 1000 document(s) imported successfully. 0 document(s) failed to import.
```

- ✓ Dans MongoDB Shell ,connectez-vous à la base de données BankDB et afficher son contenu

```
PS C:\Users\oussa\Documents> mongosh "mongodb+srv://cluster0.rwjga68.mongodb.net/BankDB" --username oukhmir_oussama
Enter password: *****
Current Mongosh Log ID: 6949ae3557aed587f01e2620
Connecting to:      mongodb+srv://<credentials>@cluster0.rwjga68.mongodb.net/BankDB?appName=mongosh+2.5.10
Using MongoDB:      8.0.17
Using Mongosh:      2.5.10

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

Atlas atlas-jh69i3-shard-0 [primary] BankDB> use BankDB
already on db BankDB
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.count()
DeprecationWarning: Collection.count() is deprecated. Use countDocuments or estimatedDocumentCount.
1000
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.find().pretty()
[
  {
    _id: ObjectId('6949ad657d6782d09c255372'),
    'Transaction ID': 'TXN9412011085',
    'Sender Account ID': 'ACC58958',
    'Receiver Account ID': 'ACC32826',
    'Transaction Amount': 529.62,
    'Transaction Type': 'Withdrawal',
    'Timestamp': '2025-01-17 10:51:00',
    'Transaction Status': 'Success',
    'Fraud Flag': 'False',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice2',
    'Latency (ms)': 11,
    'Slice Bandwidth (Mbps)': 89,
    'PIN Code': 2369,
    Geolocation: { type: 'Point', coordinates: [ 118.2437, 35.6895 ] }
  },
  {
    _id: ObjectId('6949ad657d6782d09c255373'),
    'Transaction ID': 'TXN6682052855',
    'Sender Account ID': 'ACC68764',
    'Receiver Account ID': 'ACC53807',
    'Transaction Amount': 1171.23,
    'Transaction Type': 'Transfer',
    'Timestamp': '2025-01-17 10:29:00',
    Geolocation: { type: 'Point', coordinates: [ -37.6173, 35.6895 ] }
  },
  {
    _id: ObjectId('6949ad657d6782d09c255384'),
    'Transaction ID': 'TXN3914566506',
    'Sender Account ID': 'ACC45926',
    'Receiver Account ID': 'ACC76590',
    'Transaction Amount': 979.89,
    'Transaction Type': 'Withdrawal',
    'Timestamp': '2025-01-17 10:25:00',
    'Transaction Status': 'Failed',
    'Fraud Flag': 'False',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice2',
    'Latency (ms)': 15,
    'Slice Bandwidth (Mbps)': 225,
    'PIN Code': 7119,
    Geolocation: { type: 'Point', coordinates: [ -2.3522, 40.7128 ] }
  },
  {
    _id: ObjectId('6949ad657d6782d09c255385'),
    'Transaction ID': 'TXN6302601632',
    'Sender Account ID': 'ACC91042',
    'Receiver Account ID': 'ACC98945',
    'Transaction Amount': 129.78,
    'Transaction Type': 'Transfer',
    'Timestamp': '2025-01-17 10:11:00',
    'Transaction Status': 'Success',
    'Fraud Flag': 'True',
    'Device Used': 'Desktop',
    'Network Slice ID': 'Slice1',
    'Latency (ms)': 19,
    'Slice Bandwidth (Mbps)': 106,
    'PIN Code': 5199,
    Geolocation: { type: 'Point', coordinates: [ 118.2437, 51.5074 ] }
  }
]
Type "it" for more
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

```
mongosh mongodb+srv://<credentials>@cluster0.rwjga68.mongodb.net/BankDB?appName=mongosh+2.5.10
Atlas atlas-jh69i3-shard-0 [primary] BankDB> use BankDB
already on db BankDB
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.find().pretty()
[
  {
    _id: ObjectId('6949ad657d6782d09c255384'),
    'Transaction ID': 'TXN3914566506',
    'Sender Account ID': 'ACC45926',
    'Receiver Account ID': 'ACC76590',
    'Transaction Amount': 979.89,
    'Transaction Type': 'Withdrawal',
    'Timestamp': '2025-01-17 10:25:00',
    'Transaction Status': 'Failed',
    'Fraud Flag': 'False',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice2',
    'Latency (ms)': 15,
    'Slice Bandwidth (Mbps)': 225,
    'PIN Code': 7119,
    Geolocation: { type: 'Point', coordinates: [ -2.3522, 40.7128 ] }
  },
  {
    _id: ObjectId('6949ad657d6782d09c255385'),
    'Transaction ID': 'TXN6302601632',
    'Sender Account ID': 'ACC91042',
    'Receiver Account ID': 'ACC98945',
    'Transaction Amount': 129.78,
    'Transaction Type': 'Transfer',
    'Timestamp': '2025-01-17 10:11:00',
    'Transaction Status': 'Success',
    'Fraud Flag': 'True',
    'Device Used': 'Desktop',
    'Network Slice ID': 'Slice1',
    'Latency (ms)': 19,
    'Slice Bandwidth (Mbps)': 106,
    'PIN Code': 5199,
    Geolocation: { type: 'Point', coordinates: [ 118.2437, 51.5074 ] }
  }
]
Type "it" for more
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

- ✓ Affiche les collections de la base de données courante.
- ✓ Afficher les informations sur une transaction pour comprendre la structure
- ✓ Combien y a-t-il de transactions dans la base de données ?
- ✓ Trouver toutes les transactions échouées ("transaction status": "failed")
- ✓ Vérifier les transactions suspectes ("fraud flag" : "true")

```

mongosh mongodb+srv://ca x + v
Atlas atlas-jh69i3-shard-0 [primary] BankDB> show collections
transactions
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.findOne()
{
  _id: ObjectId('6949ad657d6782d09c255372'),
  'Transaction ID': 'TXN9412011085',
  'Sender Account ID': 'ACC58958',
  'Receiver Account ID': 'ACC32826',
  'Transaction Amount': 529.62,
  'Transaction Type': 'Withdrawal',
  'Timestamp': '2025-01-17 10:51:00',
  'Transaction Status': 'Success',
  'Fraud Flag': 'False',
  'Device Used': 'Mobile',
  'Network Slice ID': 'Slice2',
  'Latency (ms)': 11,
  'Slice Bandwidth (Mbps)': 89,
  'PIN Code': 2369,
  'Geolocation': { type: 'Point', coordinates: [ 118.2437, 35.6895 ] }
}
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.countDocuments()
1000
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |

mongosh mongodb+srv://ca x + v
1000
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.find({ "Transaction Status": "Failed" }).pretty()
[
  {
    _id: ObjectId('6949ad657d6782d09c255374'),
    'Transaction ID': 'TXN4351744832',
    'Sender Account ID': 'ACC55704',
    'Receiver Account ID': 'ACC24911',
    'Transaction Amount': 817.55,
    'Transaction Type': 'Deposit',
    'Timestamp': '2025-01-17 10:13:00',
    'Transaction Status': 'Failed',
    'Fraud Flag': 'False',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice2',
    'Latency (ms)': 8,
    'Slice Bandwidth (Mbps)': 82,
    'PIN Code': 2092,
    'Geolocation': { type: 'Point', coordinates: [ -2.3522, 40.7128 ] }
  },
  {
    _id: ObjectId('6949ad657d6782d09c255375'),
    'Transaction ID': 'TXN9743166792',
    'Sender Account ID': 'ACC82066',
    'Receiver Account ID': 'ACC33985',
    'Transaction Amount': 579.65,
    'Transaction Type': 'Transfer',
    'Timestamp': '2025-01-17 10:17:00',
    'Transaction Status': 'Failed',
    'Fraud Flag': 'False',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice2',
    'Latency (ms)': 14,
    'Slice Bandwidth (Mbps)': 246,
    'PIN Code': 9680,
    'Geolocation': { type: 'Point', coordinates: [ -0.1278, 51.5074 ] }
  },
  {
    _id: ObjectId('6949ad657d6782d09c255377'),
    'Transaction ID': 'TXN3008400720',
    'Sender Account ID': 'ACC96801',
  }
]

```



```

mongosh mongodb+srv://<ci> x + v
'Receiver Account ID': 'ACC27800',
'Transaction Amount': 485.37,
'Transaction Type': 'Transfer',
'Timestamp': '2025-01-17 10:50:00',
'Transaction Status': 'Failed',
'Fraud Flag': 'True',
'Device Used': 'Mobile',
'Network Slice ID': 'Slice3',
'Latency (ms)': 20,
'Slice Bandwidth (Mbps)': 173,
'PIN Code': 9631,
'Geolocation': { type: 'Point', coordinates: [ -37.6173, 34.0522 ] }
},
{
  _id: ObjectId('6949ad657d6782d09c255391'),
  'Transaction ID': 'TXN9520068950',
  'Sender Account ID': 'ACC14994',
  'Receiver Account ID': 'ACC16656',
  'Transaction Amount': 495.0,
  'Transaction Type': 'Deposit',
  'Timestamp': '2025-01-17 10:14:00',
  'Transaction Status': 'Failed',
  'Fraud Flag': 'True',
  'Device Used': 'Desktop',
  'Network Slice ID': 'Slice3',
  'Latency (ms)': 10,
  'Slice Bandwidth (Mbps)': 170,
  'PIN Code': 3075,
  'Geolocation': { type: 'Point', coordinates: [ 74.006, 34.0522 ] }
}
]
Type "it" for more
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.find({ "Fraud Flag": true }).pretty()
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.find({ "Fraud Flag": true }).pretty()
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.countDocuments({ "Fraud Flag": true })
0
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.countDocuments({ "Transaction Status": "Failed" })
513
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |

```

✓ Nombre total de transactions par statut

```

mongosh mongodb+srv://<ci> x + v
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.aggregate([
... { $group: { _id: "$Transaction Status", total: { $count: {} } } }
... ])
[ { _id: 'Failed', total: 513 }, { _id: 'Success', total: 487 } ]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |

```

✓ Montant moyen des transactions par type

```

mongosh mongodb+srv://<ci> x + v
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.aggregate([
... { $group: { _id: "$Transaction Type", avgAmount: { $avg: "$Transaction Amount" } } }
... ])
[
  { _id: 'Transfer', avgAmount: 780.1512032085561 },
  { _id: 'Deposit', avgAmount: 797.6032278481013 },
  { _id: 'Withdrawal', avgAmount: 733.3745806451612 }
]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |

```

✓ Top 5 des comptes qui envoient le plus d'argent

```

mongosh mongodb+srv://<ci> x + v
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.aggregate([
... { $group: { _id: "$Sender Account ID", totalSent: { $sum: "$Transaction Amount" } } }, { $sort: { totalSent: -1 } },
... { $limit: 5 }
... ])
[
  { _id: 'ACC37810', totalSent: 2757.77 },
  { _id: 'ACC75741', totalSent: 1918.82 },
  { _id: 'ACC89865', totalSent: 1692.55 },
  { _id: 'ACC44804', totalSent: 1497.76 },
  { _id: 'ACC67128', totalSent: 1495.01 }
]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |

```

✓ Calculer le montant total des transactions par type

```
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.aggregate([
... { $group: { _id: "$Transaction Type", total: { $sum: "$Transaction Amount" } } }
... ])
[
  { _id: 'Withdrawal', total: 227346.12 },
  { _id: 'Transfer', total: 291776.55 },
  { _id: 'Deposit', total: 252042.62 }
]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

✓ Vérifier la latence moyenne par "Network Slice ID"

```
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.aggregate([
... { $group: { _id: "$Network Slice ID", avgLatency: { $avg: "$Latency (ms)" } } }
... ])
[
  { _id: 'Slice2', avgLatency: 11.3 },
  { _id: 'Slice1', avgLatency: 11.91640866873065 },
  { _id: 'Slice3', avgLatency: 11.86053412462908 }
]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

✓ Transactions échouées par appareil utilisé

```
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.aggregate([
... { $match: { "Transaction Status": "Failed" } }
... { $group: { _id: "$Device Used", totalFailed: { $count: {} } } }
... ])
[
  { _id: 'Desktop', totalFailed: 243 },
  { _id: 'Mobile', totalFailed: 270 }
]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

✓ Trouver les transactions suspectes (fraude et montant élevé > 1000)

```
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.find({ "Fraud Flag": "True", "Transaction Amount": { $gt: 1000 } })
[
  {
    _id: ObjectId('6949ad657d6782d09c25537e'),
    'Transaction ID': 'TXN3109277527',
    'Sender Account ID': 'ACC48227',
    'Receiver Account ID': 'ACC93536',
    'Transaction Amount': 1135.8,
    'Transaction Type': 'Deposit',
    'Timestamp': '2025-01-17 10:56:00',
    'Transaction Status': 'Failed',
    'Fraud Flag': 'True',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice2',
    'Latency (ms)': 11,
    'Slice Bandwidth (Mbps)': 69,
    'PIN Code': 8607,
    'Geolocation': { type: 'Point', coordinates: [ 118.2437, 48.8566 ] }
  },
  {
    _id: ObjectId('6949ad657d6782d09c255383'),
    'Transaction ID': 'TXN7417651729',
    'Sender Account ID': 'ACC26106',
    'Receiver Account ID': 'ACC21043',
    'Transaction Amount': 1371.8,
    'Transaction Type': 'Transfer',
    'Timestamp': '2025-01-17 10:09:00',
    'Transaction Status': 'Failed',
    'Fraud Flag': 'True',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice1',
    'Latency (ms)': 8,
    'Slice Bandwidth (Mbps)': 61,
    'PIN Code': 3326,
    'Geolocation': { type: 'Point', coordinates: [ -37.6173, 35.6895 ] }
  },
  {
    _id: ObjectId('6949ad657d6782d09c25538c'),
    'Transaction ID': 'TXN2214150284',
    'Sender Account ID': 'ACC48650',
    'Receiver Account ID': 'ACC48650',
    'Transaction Amount': 1000.0,
    'Transaction Type': 'Transfer',
    'Timestamp': '2025-01-17 10:09:00',
    'Transaction Status': 'Failed',
    'Fraud Flag': 'True',
    'Device Used': 'Mobile',
    'Network Slice ID': 'Slice1',
    'Latency (ms)': 8,
    'Slice Bandwidth (Mbps)': 61,
    'PIN Code': 3326,
    'Geolocation': { type: 'Point', coordinates: [ -37.6173, 35.6895 ] }
  }
]
```


- ✓ Combien transactions ont un montant entre 100 et 200 ans ?

```
mongosh mongodb+srv://<ci> x + v
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.countDocuments({ amount: { $gte: 100, $lte: 200 } })
0
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

- ✓ Afficher Montant total des transaction

```
mongosh mongodb+srv://<ci> x + v
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.aggregate([{$group: { _id: null, totalAmount: { $sum: "$Transaction Amount" } } }])
[ { _id: null, totalAmount: 771165.29 } ]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

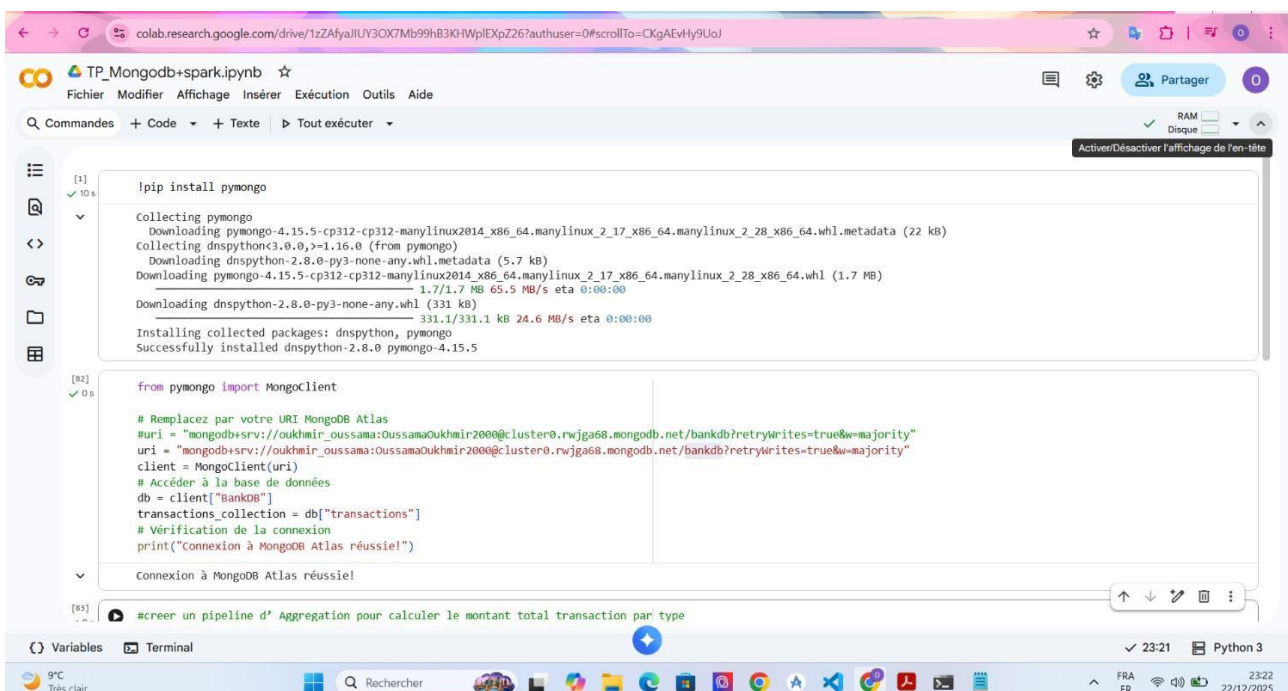
- ✓ afficher Taux d'échec des transactions (%) et Afficher le Nombre de transactions frauduleuses

```
mongosh mongodb+srv://<ci> x + v
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.aggregate([
... { $group: { _id: "$Transaction Status", count: { $count: {} } } }
... ])
[ { _id: 'Success', count: 487 }, { _id: 'Failed', count: 513 } ]
Atlas atlas-jh69i3-shard-0 [primary] BankDB> db.transactions.countDocuments({ "Fraud Flag": "True" })
481
Atlas atlas-jh69i3-shard-0 [primary] BankDB> |
```

6. Manipuler MapReduce sur la collection transactions

- ✓ Exécutez la cellule suivante dans Google Colab pour installer PyMongo :

!pip install pymongo



```
!pip install pymongo

Collecting pymongo
  Downloading pymongo-4.15.5-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (22 kB)
Collecting dnspython<3.0.0,>=1.16.0 (from pymongo)
  Downloading dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
Downloading pymongo-4.15.5-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (1.7 MB)
  1.7/1.7 MB 65.5 MB/s eta 0:00:00
Downloading dnspython-2.8.0-py3-none-any.whl (331 kB)
  331.1/331.1 kB 24.6 MB/s eta 0:00:00
Installing collected packages: dnspython, pymongo
Successfully installed dnspython-2.8.0 pymongo-4.15.5

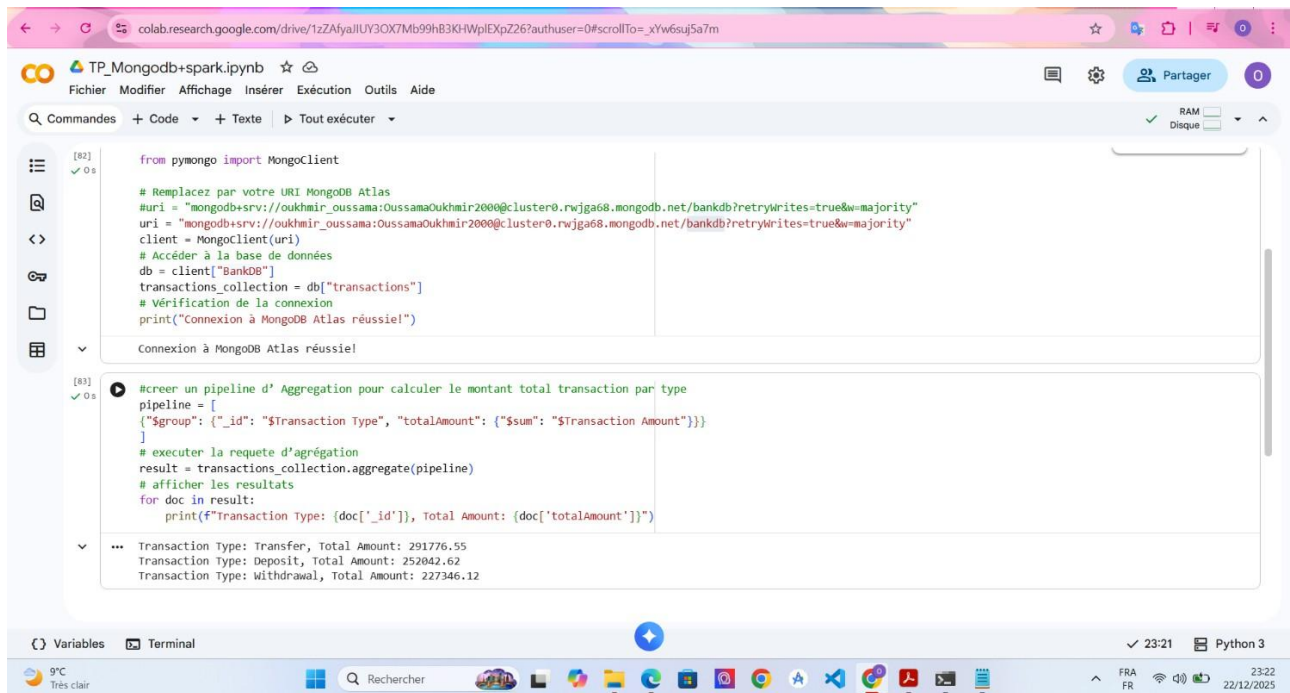
from pymongo import MongoClient

# Remplacez par votre URI MongoDB Atlas
#uri = "mongodb+srv://oukhir_oussama:OussamaOukhir2000@cluster0.rwjga68.mongodb.net/bankdb?retryWrites=true&w=majority"
uri = "mongodb+srv://oukhir_oussama:OussamaOukhir2000@cluster0.rwjga68.mongodb.net/bankdb?retryWrites=true&w=majority"
client = MongoClient(uri)
# Accéder à la base de données
db = client["bankdb"]
transactions_collection = db["transactions"]
# Vérification de la connexion
print("Connexion à MongoDB Atlas réussie!")

Connexion à MongoDB Atlas réussie!

#créer un pipeline d' Aggregation pour calculer le montant total transaction par type
```

- ✓ Ensuite, Se connecter à MongoDB Atlas et Etant donné que map_reduce() est obsolète, utilisez le framework d'agrégation de MongoDB pour additionner les montants des transactions par type de transaction



The screenshot shows a Google Colab notebook interface. The notebook title is "TP_Mongodb+spark.ipynb". The code is written in Python and consists of two main parts: connecting to MongoDB Atlas and performing an aggregation query.

```
[82] ✓ 0 s
from pymongo import MongoClient

# Remplacez par votre URI MongoDB Atlas
#uri = "mongodb+srv://oukhir_oussama:OussamaOukhir2000@cluster0.rwjga68.mongodb.net/bankdb?retryWrites=true&w=majority"
uri = "mongodb+srv://oukhir_oussama:OussamaOukhir2000@cluster0.rwjga68.mongodb.net/bankdb?retryWrites=true&w=majority"
client = MongoClient(uri)
# Accéder à la base de données
db = client["bankdb"]
transactions_collection = db["transactions"]
# Vérification de la connexion
print("Connexion à MongoDB Atlas réussie!")

Connexion à MongoDB Atlas réussie!

[83] ✓ 0 s
#créer un pipeline d' Agregation pour calculer le montant total transaction par type
pipeline = [
    {"$group": {"_id": "$Transaction Type", "totalAmount": {"$sum": "$Transaction Amount"}}}
]
# executer la requete d'agregation
result = transactions_collection.aggregate(pipeline)
# afficher les resultats
for doc in result:
    print(f"Transaction Type: {doc['_id']}, Total Amount: {doc['totalAmount']}")

...
Transaction Type: Transfer, Total Amount: 291776.55
Transaction Type: Deposit, Total Amount: 252842.62
Transaction Type: Withdrawal, Total Amount: 227346.12
```

The bottom of the notebook shows the "Variables" and "Terminal" tabs. The "Variables" tab is active, showing a list of variables. The "Terminal" tab is also visible. The system tray at the bottom indicates the temperature is 9°C, the weather is "Très clair", and the time is 23:21 on 22/12/2025.

7. Dashboard avec MongoDB Atlas

<https://charts.mongodb.com/charts-project-0-ixdjewa/public/dashboards/6G4Gc5a0-2d7b-48bf-8bc7-43dee00a4c00>

